

TEAM ID	SWTID-2026-8827
PROJECT TITLE	India's Agricultural Crop Production Analysis.
MAX MARKS	5 MARKS

CHAPTER – VII

FUNCTIONAL AND PERFORMANCE TESTING

7.1 – Performance Testing & Functional Testing

1.Amount of Data Rendered to Tableau

The project utilizes a comprehensive agricultural dataset containing crop production information from various states and districts of India. The dataset includes attributes such as State Name, District Name, Crop Year, Season, Crop Type, Area Cultivated, Production Quantity, and Yield. Approximately 20,000–30,000 records are imported into Tableau for analysis, enabling detailed visualization of agricultural trends across different regions and years.

2. Utilization of Filters

Interactive filters are implemented in Tableau to allow users to explore data efficiently.

The major filters used include:

- State Filter – View crop production for a specific state.
- District Filter – Analyze district-wise production.
- Crop Filter – Select individual crops for detailed analysis.
- Year Filter – Compare production across different years.
- Season Filter – Examine seasonal crop performance.

- Production Range Filter – Focus on specific production levels.

These filters help users perform dynamic and customized analysis.

3. Number of Calculation Fields

The dashboard uses several calculated fields to derive meaningful insights from raw data. Examples include:

1. Total Production
2. Average Yield
3. Production Growth Rate (%)
4. Area Utilization Ratio
5. Yield per Hectare
6. State-wise Production Percentage
7. Year-over-Year Production Change
8. Ranking of States by Production

Total Calculation Fields Used: 8–10

4. Number of Visualizations / Graphs

The Tableau dashboard contains multiple visualizations to present agricultural data effectively:

1. KPI Card – Total Production
2. KPI Card – Total Yield
3. Bar Chart – Top Producing States
4. Line Chart – Production Trend by Year
5. Pie Chart – Crop-wise Production Share
6. Map Visualization – State-wise Production Distribution
7. Tree Map – Crop Production Comparison

8. Stacked Bar Chart – Seasonal Production Analysis
9. Area Chart – Production Growth Trend
10. Dashboard Summary Table

Total Visualizations/Graphs: 8–10

These components collectively create an interactive and insightful dashboard for analyzing India's agricultural crop production patterns.

